

Light Up RGB Spotlight

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[Feature Description](#)

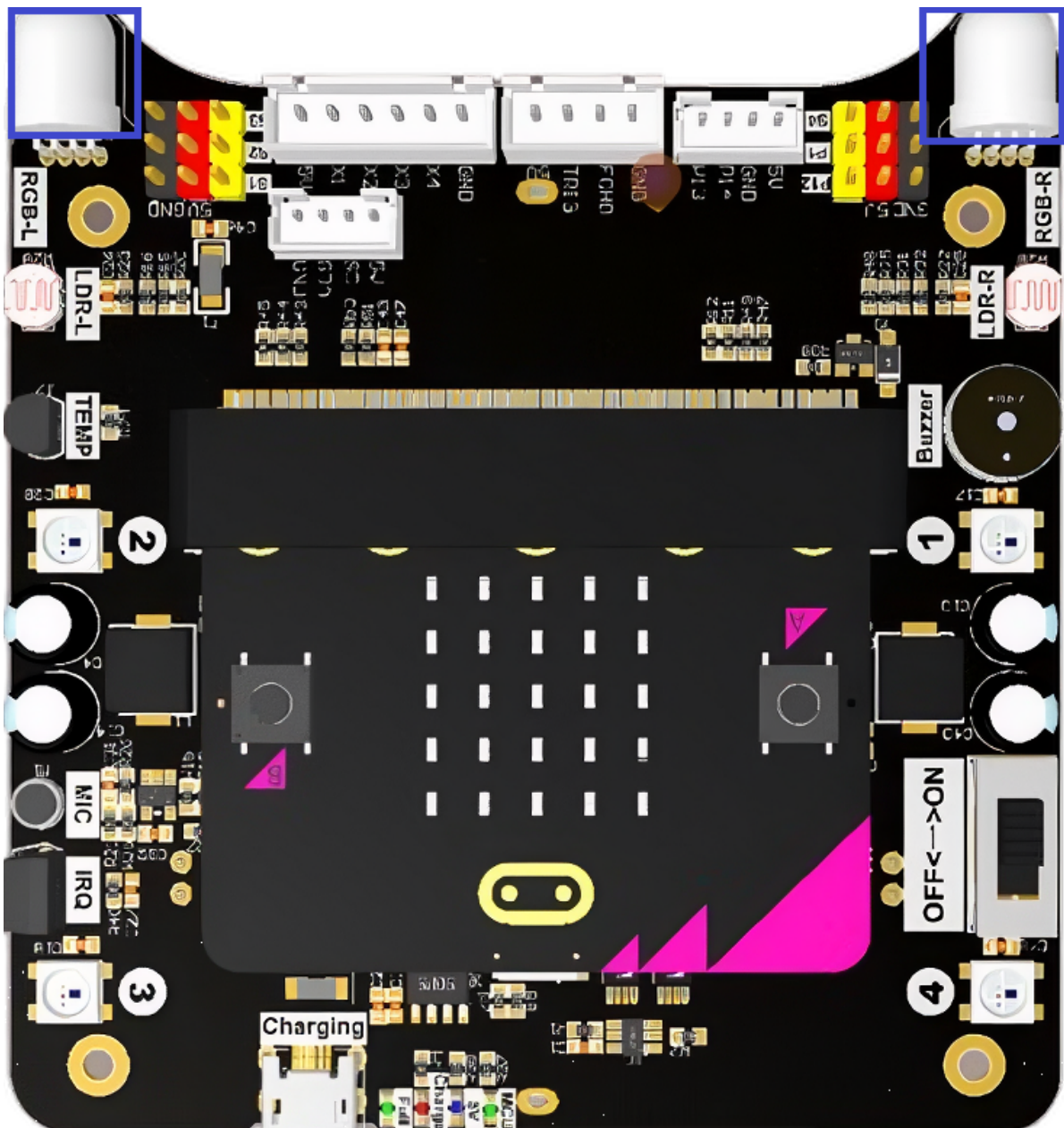
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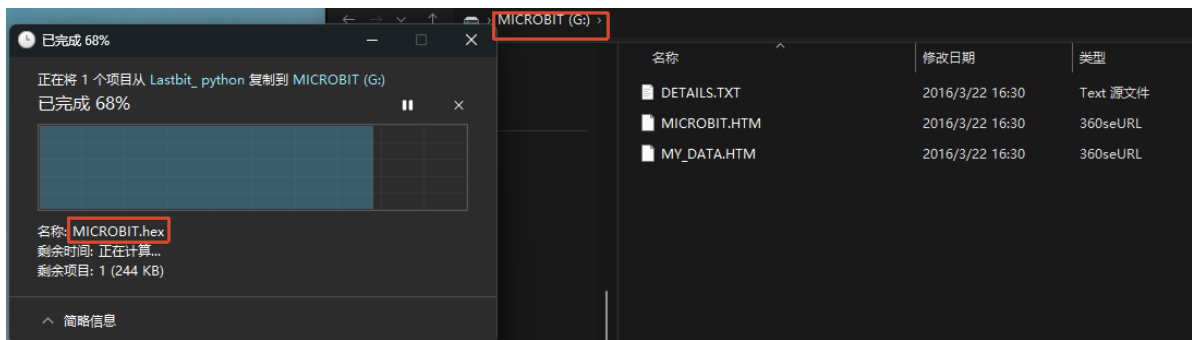
Feature Description

In this section, we will learn how to light up the RGB spotlight.



Experimental Steps

Before flashing, make sure `LASTBIT.hex` has been flashed; otherwise, an error will occur and the module cannot be imported.



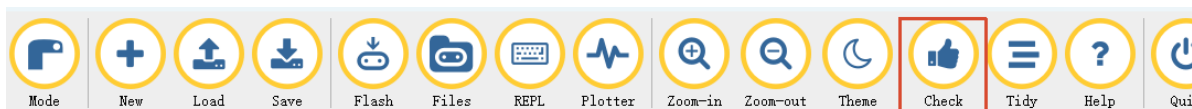
1. Before flashing, switch the switch to the OFF position and connect the computer to the microbit main board using a microUSB cable. **Note: it should be the Microbit interface, not the expansion board's Charging port.**
2. Open the Mu software. Here you can directly copy the program source code we provide; the operation steps are as follows. You can also enter the code yourself in the editor window. Note! All English characters and symbols should be entered in English input mode. Use the Tab key for indentation, and end the last line with a blank line.
Click the **New** button in the toolbar at the top left, and a file titled untitled will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a small red dot after the title bar, indicating that the code content has been modified but is not yet saved.

Whether or not you save does not affect the code check and flashing.

4. At this point, you can click **check** to check whether the code format has any problems. The prompt below tells us that the source code has no syntax errors.

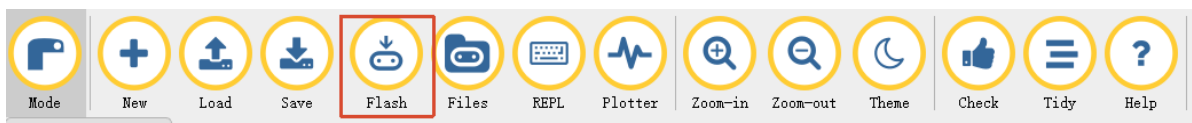


```
untitled X
7 Lastbit = LASTBIT()
8
9 # 关闭所有灯 Close all lights
10 Lastbit.all_lights_off()
11
12 while True:
13     # 红色 Red light
14     Lastbit.set_headlight_both('RED')
15     sleep(2000)
16
17     # 绿色 Green light
18     Lastbit.set_headlight_both((0, 255, 0))
19     sleep(2000)
20
21     # 蓝色 Blue light
22     # Lastbit.set_headlight_left_right('BLUE', 'BLUE')
23     # 也可以直接设置为RGB值
24     Lastbit.set_headlight_left_right((0, 0, 255), (0, 0, 255))
25     sleep(2000)
26
27     # 黄色 Yellow light
28     Lastbit.set_headlight_both('YELLOW')
29     sleep(2000)
30     # 关闭灯光 Close the light
31     Lastbit.set_headlight_both('CLOSE')
32     sleep(2000)
33
Hurraht! Checker turned up no problems.
```

5. If you have connected the MICROBIT to the PC in advance following the steps above and have flashed `LASTBIT.hex`, the next step is to download the current program to the Microbit.

Click the `Flash` tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently, and the editor will show `Copied code onto micro:bit.` at the bottom, indicating that the flashing is complete. After flashing, the indicator light will no longer blink.



6. Switch the switch to the ON position.

Code Implementation

```
# 1.Light_Up_RGB_Spotlight
# Import the microbit and lastbit module
from microbit import *
# Import the lastbit module
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Close all lights
Lastbit.all_lights_off()
```

```

while True:
    # Red light
    Lastbit.set_headlight_both('RED')
    sleep(2000)

    # Green light
    Lastbit.set_headlight_both((0, 255, 0))
    sleep(2000)

    # Blue light
    # Lastbit.set_headlight_left_right('BLUE', 'BLUE')
    # Can also be set directly via RGB values
    Lastbit.set_headlight_left_right((0, 0, 255), (0, 0, 255))
    sleep(2000)

    # Yellow light
    Lastbit.set_headlight_both('YELLOW')
    sleep(2000)
    # Close the light
    Lastbit.set_headlight_both('CLOSE')
    sleep(2000)

```

`set_headlight_both()` lights up the spotlight and supports 9 predefined colors. This method is used to control the left and right spotlights to the same color at the same time. It also supports passing in a color tuple (R, G, B), for example (0, 255, 0) means green.

- Red: 'RED'
- Orange: 'ORANGE'
- Yellow: 'YELLOW'
- Green: 'GREEN'
- Cyan: 'CYAN'
- Blue: 'BLUE'
- Purple: 'PURPLE'
- White: 'WHITE'
- Black (off): 'CLOSE'

The `set_headlight_left_right(left, right)` method supports setting the left and right lights to different colors. It also supports the 9 colors above and accepts a color tuple (R, G, B), for example (0, 255, 0) means green. The first parameter is the color of the left light, and the second parameter is the color of the right light.

Experimental Phenomenon

After flashing the program, the spotlights at the front of the car will light up in a cycle of red -> green -> blue -> yellow, switching the light effect every 2 seconds.